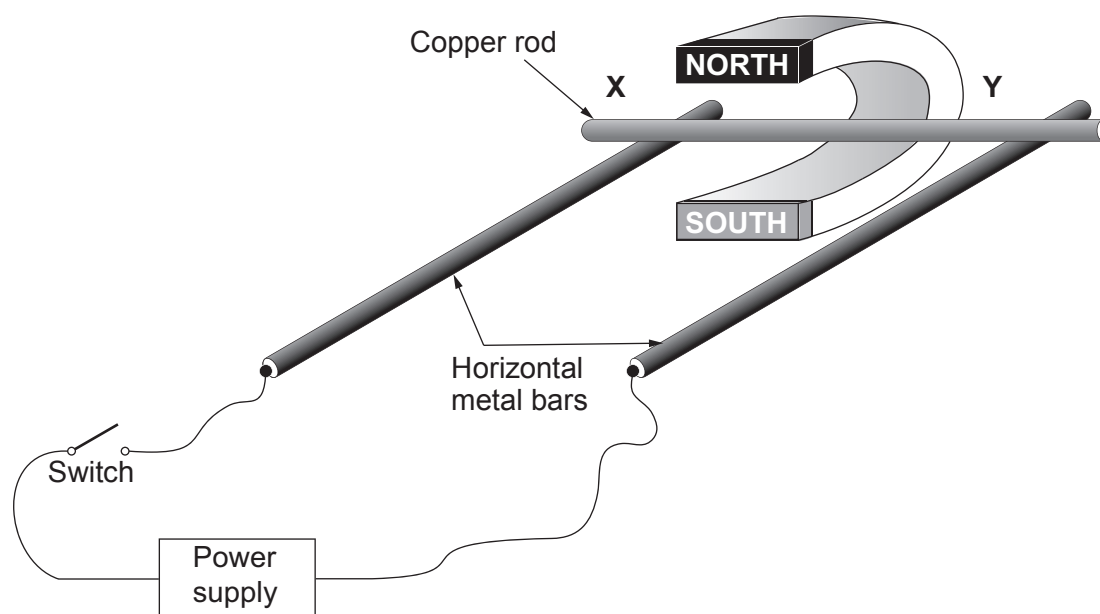


4. The following experiment is set up in a school laboratory for students.



- (a) (i) Explain why, when the switch in the circuit is closed, the copper rod starts to roll along the two horizontal metal bars. [2]

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- (ii) When the switch is closed the copper rod moves away from the magnet. One of the students concludes that the direction of the current is from X to Y. Explain if you agree or disagree with their conclusion. [2]

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- (iii) State **two** changes that could be made to the apparatus so that the copper rod moves faster when the switch is closed. [2]

1.
2.

- (b) The effect observed in this experiment is used in many everyday devices. Give an example of **one** such use. [1]

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